COSC 51033 (Badge 04)

Advanced Operating System

Assignment 03

Consider the following five processes P1, P2, P3, P4 and P5, with the arrival time and the length of the CPU burst given in milliseconds.

Process ID	Arrival Time	Burst Time
P1	0	12
P2	2	10
P3	4	5
P4	6	16
P5	10	8

Consider the following three process scheduling algorithms.

(i) First Come First Serve (FCFS) algorithm
(ii) Shortest Job First (SJF) algorithm (Using Preemptive behavior)
(iii) Round Robin (RR) algorithm (Using the time quantum of 5 milliseconds and Preemptive behavior)

1. Draw a Gantt chart (Timeline) for each of the above three scheduling algorithms.

2. Compute the **Waiting time and Turnaround time for every process** for each of the above three scheduling algorithms.

3. Compute **Average waiting time and Average turnaround time** for each of the above three scheduling algorithms.

4. Prepare the average values in a table for easy comparison. Which algorithm do you think best is best suited to execute the above processors.

Submit on or before 2026.04.23